Number Theory — problem 6.138

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(a) Let a_n be the last nonzero digit in the decimal representation of the number $n!$. Does the sequence $a_1, a_2, a_3, \ldots$ become periodic after a finite number of steps (USSR proposal for IMO 1991) ?

(b) Let d_n be the last nonzero digit of $n!$. Prove that d_n is not periodic, that is, p and n_0 do not exist such that $d_{n+p} = d_n$ for all $n \geq n_0$ (USSR proposal for IMO 1983).